

Team name: ASTRATEK

Challenge: Meteor Madness

Project name: IMPACTOR-2025

References & Data Sources

1. Astronomical Definitions and Background

- NASA Solar System Exploration: Asteroids, Comets & Meteors — <https://solarsystem.nasa.gov/asteroids-comets-and-meteors/overview/>
- ESA Education Portal – Asteroids, Comets, and Meteoroids — https://www.esa.int/Science_Exploration/Space_Science
- NASA JPL – What's the Difference Between an Asteroid, Comet, and Meteor? — <https://www.jpl.nasa.gov/edu/learn/project/whats-the-difference-between-an-asteroid-comet-and-meteor/>

2. Missions and Space Programs

NASA Planetary Defense and NEO Missions

- NASA Planetary Defense Coordination Office — <https://cneos.jpl.nasa.gov/>
- NASA NEO Surveyor Mission Page — <https://www.nasa.gov/mission/neo-surveyor/>

ESA Planetary Defence Programs

- ESA Planetary Defence Office — https://www.esa.int/Safety_Security/Planetary_Defence
- ESA HERA Mission Overview — https://www.esa.int/Space_Safety/Hera

NASA OSIRIS-REx Mission

- OSIRIS-REx Main Mission Page — <https://www.nasa.gov/osiris-rex>
- Asteroid Bennu Data — <https://www.asteroidmission.org/>

NASA DART Mission

- DART Mission Overview — <https://dart.jhuapl.edu/>
- DART Impact Results (Dimorphos) — <https://www.nasa.gov/feature/nasa-s-dart-mission-hits-asteroid-in-first-planetary-defense-test>

3. Famous Asteroids and Impact Events

Chelyabinsk Meteor (2013)

- NASA Science – Chelyabinsk Meteor — <https://science.nasa.gov/asteroids/comets-meteors/chelyabinsk-meteor/>
- NASA JPL NEO News — <https://neo.jpl.nasa.gov/news/news179.html>

Tunguska Event (1908)

- NASA Earth Observatory – Tunguska Event — <https://earthobservatory.nasa.gov/features/Tunguska>
- ESA Space Safety Articles – Tunguska and Impact Risk — https://www.esa.int/Safety_Security

Meteor (Barringer) Crater

- USGS Impact Crater Data — <https://www.usgs.gov/>
- Meteor Crater Official Site — <https://meteorcrater.com/>

4. Asteroid and Meteor Classification

- NASA Asteroid Types Overview — <https://solarsystem.nasa.gov/asteroids-comets-and-meteors/asteroids/in-depth/>
- ESA Space Science Portal – Asteroid Taxonomy — https://www.esa.int/Science_Exploration/Space_Science/Asteroids

5. Theoretical Physics and Equations

- Vallado, D. A. (2013). *Fundamentals of Astrodynamics and Applications* (4th ed.). Microcosm Press.
- NASA Glenn Research Center – Aerodynamics Learning Portal — <https://www.grc.nasa.gov/www/k-12/airplane/>
- NASA Ames Research Center – Atmospheric Entry and Ablation Models — <https://ntrs.nasa.gov/citations/20180004325>
- NASA SpacePlace – Kepler’s Laws and Orbits — <https://spaceplace.nasa.gov/kepler/en/>

6. Atmospheric and Material Data

- U.S. Standard Atmosphere (1976) — NASA and NOAA Joint Publication.
- NASA Thermophysical Properties Database — (Material heat and ablation data, internal reference).
- NASA Technical Reports Server (NTRS) — Impact and Fragmentation Modeling Reports — <https://ntrs.nasa.gov/>

7. USGS and NOAA Earth Data

- USGS Earthquake Catalog — <https://earthquake.usgs.gov/earthquakes/search/>
- USGS The National Map — <https://www.usgs.gov/programs/national-geospatial-program/national-map>
- USGS ShakeMap Program — <https://earthquake.usgs.gov/data/shakemap/>
- NOAA ETOPO1 Global Relief Model — <https://www.ncei.noaa.gov/products/etopo-global-relief-model>

8. NASA / JPL Small-Body and NEO Databases

- NASA CNEOS Fireball & Bolide Database — <https://cneos.jpl.nasa.gov/fireballs/>
- NASA CNEOS Close-Approach Database — <https://cneos.jpl.nasa.gov/ca/>
- NASA / JPL Small-Body Database (SBDB) Query Tool — https://ssd.jpl.nasa.gov/tools/sbdb_query.html
- NASA SEDAC Gridded Population of the World (GPWv4) — <https://sedac.ciesin.columbia.edu/data/collection/gpw-v4>

9. Scientific and Modeling References

- Melosh, H. J. (1989). *Impact Cratering: A Geologic Process*. Oxford University Press.
- Collins, G. S., Melosh, H. J., & Marcus, R. A. (2005). *Earth Impact Effects Program. Meteoritics & Planetary Science*, 40(6).
- Ward, S. N., & Asphaug, E. (2000). *Asteroid Impact Tsunami: A Probabilistic Hazard Assessment. Icarus*, 145.
- Boslough, M. B. E., & Crawford, D. A. (2008). *Low-Altitude Airbursts and the Impact Threat. Int. J. Impact Eng.*, 35(12).
- Popova, O. P., et al. (2013). *Chelyabinsk Airburst, Damage and Meteorite Recovery. Science*, 342.
- Toon, O. B., Zahnle, K., Morrison, D., Turco, R. P., & Covey, C. (1997). *Environmental Perturbations Caused by the Impacts of Asteroids and Comets. Reviews of Geophysics*, 35(1).
- Kring, D. A. (2007). *The Chicxulub Impact Event and Its Environmental Consequences. Palaeogeography, Palaeoclimatology, Palaeoecology*, 255, 4–21.
- Lambeck, K. (1980). *The Earth's Variable Rotation*. Cambridge University Press.

10. Acknowledgment

This work uses open-access data and mission information provided by:

- NASA (JPL, CNEOS, SEDAC, Ames, Glenn Research Center)
- USGS (Earthquake Hazards Program, The National Map, ShakeMap)

- NOAA (National Centers for Environmental Information)
- ESA (Planetary Defence Office, HERA Program)

All data are used in accordance with open-data policies for research and educational purposes under the NASA Space Apps Challenge 2025 framework.